## Recent paper statements

These are adapted statements with explicit Lean definitions. Their translation has been checked; the three paper results have not been proved in Lean.

For the magnetic identity, $M(A, q) = e^{2\pi iq}A + e^{-2\pi iq}A^{\mathsf{T}}$. The profile $\Phi_A(m, k)$ sums ordered products with $k$ forward factors $A$ and $m - k$ transposed factors. Its empty product is the identity. We define

$$\text{walkFourierSum}(A, m, q) = e^{-2\pi iqm} \sum_{k=0}^{m} e^{4\pi iqk} \Phi_A(m, k).$$

For a graph, $\phi(G, \mu)$ minimizes cut conductance over sets whose volume lies between $\mu \operatorname{vol}(G)$ and $(1-\mu)\operatorname{vol}(G)$, with both sides positive; $\phi_0(G) = \phi(G, 0)$. The value $\lambda(G, \mu)$ is the infimum of $x^{\mathsf{T}}Lx$ subject to

$$x^{\mathsf{T}}d = 0, \quad x^{\mathsf{T}}Dx = 1, \quad \sqrt{\frac{\mu}{(1-\mu)\operatorname{vol}(G)}} \le |x_i| \le \sqrt{\frac{1-\mu}{\mu \operatorname{vol}(G)}} \quad \text{for every } i.$$

For the SDP, $B$ supplies symmetric coefficient matrices, $\mathcal{A}_B(X)_k = \langle B_k, X \rangle$, and $\mathcal{A}_B^*(y) = \sum_k y_k B_k$. Feasibility means positive semidefiniteness, trace at most $\alpha$, and $\mathcal{A}_B(X) = b$. A minimizer is feasible and has objective no larger than every feasible matrix. The dual bound is

$$d(C, B, b, y, \alpha) = \langle y, b \rangle + \alpha \min\{\lambda_{\min}(C - \mathcal{A}_B^*(y)), 0\}.$$

**Result 1 (magnetic powers encode walk profiles).** [Huang, Gleich, Li 2025, Supplement S.II, Theorem 1]
Let $n$ be a natural number. Assume $n > 0$ (`positive`). Let $A$ be a matrix of size $n \times n$ over $\mathbb{R}$. Let $m$ be a natural number. Let $q$ be a real number. Let $M = M(A, q)$. Then

$$(M)^m = \text{walkFourierSum}(A, m, q).$$

Adapted from the cited source. Scope: forward transform.

**Result 2 (size specific Cheeger inequality).** [Huang and Gleich 2026, Theorem 3.1]
Let $n$ be a natural number. Let $G$ be a simple graph on $\{0, \ldots, n-1\}$. Let $\mu$ be a real number. Assume $G$ is connected (`connectivity`). Assume $n \geq 2$ (`nontrivial`). Assume $0 < \mu$ and $\mu \leq \frac{1}{2}$ (`range`). Assume there is an admissible cut of $G$ at size $\mu$ (`admissibility`). Let $\phi = \phi(G, \mu)$. Let $\psi = \phi_0(G)$. Let *spectral* $= \lambda(G, \mu)$. Let $t = \frac{\mu^2}{\mu^2 + \frac{1}{2} - \mu}$. Then

$$\frac{(t\,\phi + (1-t)\,\psi)^2}{4} \leq \textit{spectral} \text{ and } \textit{spectral} \leq 2\,\phi.$$

Adapted from the cited source. Scope: connected simple graphs and positive mu.

**Result 3 (trace bound gives a suboptimality certificate).** [Huang and Gleich 2026, Corollary 2, equation (3)]
Let $n$ and $m$ be natural numbers. Assume $n > 0$ and $m > 0$ (`dimensions`). Let $C$, $X$ and $Z$ be $n \times n$ matrices over $\mathbb{R}$. Let $B$ be a family of $m$ real $n \times n$ measurement matrices. Let $b$ and $y$ be $m$-vectors over $\mathbb{R}$. Let $\alpha$ be a real number. Assume $C$ is symmetric (`cost`). Assume $B$ has symmetric coefficient matrices (`measurements`). Assume $\alpha \geq 0$ (`bound`). Assume $X$ is feasible for $(B, b, \alpha)$ (`candidate`). Assume $Z$ minimizes the objective for $(C, B, b, \alpha)$ (`optimizer`). Let $d = d(C, B, b, y, \alpha)$. Then

$$\langle C, X \rangle - \langle C, Z \rangle \leq \langle C, X \rangle - d.$$

Adapted from the cited source. Scope: symmetric coefficient representation of the measurement map.